William Brower

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EDUCATION

Northwestern University | GPA: 3.18 | *Computer Engineering Major / Machine Learning & Data Science Minor*

- EA 1 - 4, PHYS 135 | Solved engineering design and physics problems using differential equations, MATLAB, and Linear Algebra for mechanical and electrical systems, optics, and electromagnetic theory.
- DTC 1 and 2 | Designed a pill box that sends daily text message reminders to the client to take their medicine. Independently wrote the C++ code that controlled these functions through an Arduino (ESP32).
- ELEC/COMP_ENG 211, 222, 303 | Digital and analog circuit analysis, design, and testing using oscilloscope, waveform visualizer, and other circuit testing devices. Created an audio visualizer, calculator, and photosensor feedback circuits. Signal processing techniques (interpolation, sampling, Fourier, Laplace, Z-Transform). Verilog, Xcelium & Genus. Logic optimization, FPGA, RTL.
- COMP_SCI 150-0, 211, 214 | Coded a Naive-Bayes categorization machine-learning project and made a functional game of checkers using Python. In C, I coded an application-style project that maps my local school area, a Python debugger using tokenization techniques, and a program that dynamically allocates memory for the debugger. Data structures, storage and time complexity and optimization, self-balancing trees.
- MATH 220, 228 | Complete multivariable vector calculus. Used Jacobians and Hessians, Green's Theorem, gradients, curls, and residues for integration techniques.
- COMM_ST 102-0 | Learned public speaking tactics and gave multiple 10+ minute public speeches.
- COMP_ENG 205 | Independently developed a pong-style game using ARM Assembly that updates based on user input. Translated code from C into Assembly—advanced knowledge of low-level data storage principles.
- ELEC_ENG 302, IEMS 313 | Statistical systems and optimization. Learned probability, distributions, Bayes, Sterling Numbers, Linear, Multi-Objective, Non-Linear, and Stochastic Optimization, and network flow.

Holy Family High School *Class of 2024 Valedictorian*
Presidential Honors, Student Ambassador, UGPA: 3.96, WGPA: 4.56, Class Rank: 1/211

EXPERIENCE

University of Colorado Anschutz Medical Campus *Research Assistant, June - August 2023*

- One of only 12 students in the state of Colorado selected to perform research at CU Anschutz. I worked full-time with three post-docs, researching chronic and polycystic kidney disease. Organized and presented key results during a poster session and documented lab progress in Excel.
- Designed a poster that outlined the basic results of the research, and used this poster in a public presentation.
- Performed statistical calculations on the data, including chi-squared tests, standard deviation, correlation coefficient (r-values), F-tests, Z-scores, T-tests, and p-values.

Teacher's Assistant *Oct. 2023 - Jun. 2024*

- Assisted US/Comparative Government teacher by creating outlines that every student used to study for tests. I used PowerPoint and Word to design materials for every class, outlining key ideas and concepts.
- Created graphs and figures to communicate complex ideas or trends in political data to students.

Step Up Youth *Volunteer, Feb. 2020 - Present*
Volunteered over 500 hours assembling and distributing concessions for a non-profit organization, where my earnings benefit underprivileged youth in the local Denver community. In addition, a portion of my earnings directly supports my college tuition. Through this, I help both my community and support my education.

ADDITIONAL INFO

Coding & Technical Skills Experience in coding with JavaScript/TypeScript, HTML, and CSS using React/Node.js, MATLAB, C++, Python, and C, alongside Linux kernel and GitHub integration into project development while using Docker containers. Knowledge of the Google Test suite for C/C++, using GDB for debugging, and various industry testing standards. Familiar with SQL querying and syntax. Knowledge of Python data processing libraries/tools (pandas, numpy, scikit-learn, matplotlib) and data fitting/categorization (k-means clustering, least-squares regression, random/decision trees, Naive Bayes, and dimensionality reduction).

Misc.

- Fluent in English, Proficient in Spanish, US Citizen, US Work Authorized, No Non-Competes
- 2024 Coca-Cola Scholar Semi-Finalist, College Board National Hispanic Scholar, AP Scholar with Distinction